

Theorem (Jensen's Inequality). *Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be convex, i.e. for every $x, y \in \mathbb{R}$ and every $t \in [0, 1]$*

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y). \quad (\text{C})$$

Let $x_1, \dots, x_n \in \mathbb{R}$ and let the weights $\lambda_1, \dots, \lambda_n$ satisfy

$$\lambda_i \geq 0 \ (i = 1, \dots, n), \quad \sum_{i=1}^n \lambda_i = 1.$$

Then

$$f\left(\sum_{i=1}^n \lambda_i x_i\right) \leq \sum_{i=1}^n \lambda_i f(x_i). \quad (\text{J})$$

Proof. We proceed by induction on the number of points n .

Base case $n = 2$. Take $\lambda_1 = t$ and $\lambda_2 = 1 - t$ with $t \in [0, 1]$. Applying (C) with $x = x_1$, $y = x_2$, $t = \lambda_1$ gives

$$f(\lambda_1 x_1 + \lambda_2 x_2) \leq \lambda_1 f(x_1) + \lambda_2 f(x_2),$$

which is precisely (J) for $n = 2$.

Reduction of zero weights. If some weight $\lambda_k = 0$, the corresponding term disappears from both sides of (J); the inequality reduces to the same statement for the remaining indices. If exactly one weight equals 1 (all others are 0), both sides of (J) equal $f(x_k)$. Hence we may assume that all weights are non-negative, at most one equals 1, and at least two are strictly positive.

Induction hypothesis. Assume that for some integer $m \geq 2$ the inequality (J) holds for every collection of at most m points with non-negative weights summing to 1. We shall prove (J) for $n = m + 1$.

Reduction to a two-point convex combination. Set

$$\alpha := \lambda_n, \quad \beta := 1 - \alpha = \sum_{i=1}^{n-1} \lambda_i \ (\beta \geq 0).$$

If $\beta = 0$ then $\alpha = 1$ and we are in the trivial case already treated; thus assume $\beta > 0$. Define the auxiliary point

$$y := \frac{1}{\beta} \sum_{i=1}^{n-1} \lambda_i x_i.$$

The coefficients λ_i/β are non-negative and sum to 1, so y is a convex combination of the first $n - 1$ points. Moreover

$$\sum_{i=1}^n \lambda_i x_i = \beta y + \alpha x_n. \quad (4.1)$$

Apply convexity to the two-point combination. Using (C) with $t = \beta$ (so $1 - t = \alpha$), $x = y$ and $y = x_n$, we obtain

$$f(\beta y + \alpha x_n) \leq \beta f(y) + \alpha f(x_n). \quad (5.1)$$

Bound $\beta f(y)$ via the induction hypothesis. Since y is a convex combination of $x_1, \dots, x_{n-1}$ with normalized weights λ_i/β , the induction hypothesis (applied with $k = n - 1 \leq m$) yields

$$f(y) \leq \sum_{i=1}^{n-1} \frac{\lambda_i}{\beta} f(x_i). \quad (6.1)$$

Multiplying (6.1) by the non-negative scalar β gives

$$\beta f(y) \leq \sum_{i=1}^{n-1} \lambda_i f(x_i). \quad (6.2)$$

Combine (5.1) and (6.2).

$$\begin{aligned} f\left(\sum_{i=1}^n \lambda_i x_i\right) &= f(\beta y + \alpha x_n) \\ &\leq \beta f(y) + \alpha f(x_n) && \text{by (5.1)} \\ &\leq \sum_{i=1}^{n-1} \lambda_i f(x_i) + \alpha f(x_n) && \text{by (6.2)} \\ &= \sum_{i=1}^n \lambda_i f(x_i), \end{aligned}$$

which is exactly (J) for $n = m + 1$.

Thus the inequality holds for $n = m + 1$. By induction it holds for every integer $n \geq 2$; the case $n = 1$ is immediate because it reduces to the identity $f(x_1) \leq f(x_1)$. $\square$

Equality case. If f is strictly convex, equality in the convexity step (5.1) can occur only when $\beta = 0$, $\alpha = 0$, or $y = x_n$. When $\beta, \alpha > 0$, the condition $y = x_n$ forces

$$x_n = \frac{1}{\beta} \sum_{i=1}^{n-1} \lambda_i x_i,$$

and the induction hypothesis shows that equality in (6.2) requires all points among $x_1, \dots, x_{n-1}$ with positive weight to coincide. Consequently, for a strictly convex f equality in (J) holds *iff* all points with positive weight are equal (the case where a single weight equals 1 is a sub-case).

For a general convex (not strictly convex) function, equality may also occur when f is affine on the convex hull of the points that carry positive weight. Hence the “*iff*” statement above is valid under the additional hypothesis of strict convexity. $\square$